

MEMS-Based Solid Propellant Microthruster Design, Simulation, Fabrication, and Testing

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Abstract—A new design concept of solid propellant microthruster is proposed for micropropulsion applications. Modeling and simulation have been performed before the fabrication of the microthrusters using MEMS technologies. At sea level, the predicted thrust magnitudes range from 0.76 mN to 4.38 mN and the estimated total impulses range from 1.16×10^{-4} N · s to 4.37×10^{-4} N · s using HTPB/AP/AL as the propellant. In space, the predicted thrust magnitudes range from 9.11 mN to 26.92 mN and the estimated total impulses range from 1.25×10^{-3} N · s to 1.70×10^{-3} N · s. Single microthruster, microthruster layers and arrays have been successfully fabricated. Preliminary testing for microcombustion is conducted to verify the feasibility of the novel design. Continuous combustion has been achieved after igniting the solid propellant and successful production of thrust has been verified by the microthruster displacement. [1006]

Index Terms—MEMS, microcombustion, micropropulsion, microthruster, solid propellant, thrust.

NOMENCLATURE

a	Temperature coefficient ($\text{kg}^{-n} \cdot \text{m}^{n+1} \cdot \text{s}^{2n-1}$).
α	Half divergence angle (Degree).
$A_c = W_c \times (\text{Trench depth})$	Chamber area (m^2).
$A_e = W_e \times (\text{Trench depth})$	Nozzle exit area (m^2).
$A_t = W_t \times (\text{Trench depth})$	Nozzle throat area (m^2).
C_p	Propellant heat capacity ($\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$).
I_t	Total impulse (N · s).
k	Thermal conductivity ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$).
K_n	Knudsen number
$\dot{m}$	Mass flow rate ($\text{kg} \cdot \text{s}^{-1}$).
n	Combustion index.
P_a	Ambient pressure (Pa).
P_c	Chamber pressure (Pa).
P_e	Nozzle exit pressure (Pa).

$$r = aP_c^n$$

T_c

T

μ

u

v

V_e

W_c

W_e

W_t

t

τ

λ

ρ

γ

Combustion rate ($\text{m} \cdot \text{s}^{-1}$).
Combustion temperature (K).

Thrust (N).

Viscosity ($\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$).

x velocity in Cartesian coordinates ($\text{m} \cdot \text{s}^{-1}$).

y velocity in Cartesian coordinates ($\text{m} \cdot \text{s}^{-1}$).

Nozzle exit velocity ($\text{m} \cdot \text{s}^{-1}$).

Chamber width (m).

Nozzle exit width (m).

Nozzle throat width (m).

Time (s).

Stress tensor ($\text{N} \cdot \text{m}^{-2}$).

Mean free path (m).

Density ($\text{kg} \cdot \text{m}^{-3}$).

Thermal capacity ratio.

I. INTRODUCTION

MEMS are forecasted to have a commercial market growth similar to its parent IC technology with diverse applications [1]. One application of MEMS technology is in the aerospace field. The development of microspacecraft can meet the requirement of reducing satellite life-cycle costs while still delivering spacecraft with the capability of performing science or commercial service. A micropropulsion system will be required in microspacecraft for station keeping, attitude control, gravitation compensation and orbit adjust.

With the utility of micropropulsion established, the ideal architecture for meeting the requirements should be assessed. There are some micropropulsion systems being investigated, such as MicroPPT, colloid microthruster, FEEP, hall microthruster, cold gas microthruster, and vaporizing microthruster [2]–[5]. The solid propellant microthruster is a relatively new class of microthruster. It requires no elaborate system of pumps and valves. Therefore, the total system complexity is minimized. Integrated with MEMS technology, the solid propellant microthruster can be specified for application in microspacecraft. Several groups are performing research on solid propellant microthrusters [6]–[9]. Their main design concepts are the three-layer sandwich configurations, which normally contain three parts consisting of a propellant combustion chamber, a micronozzle (or burst diaphragm) and an ignitor. In this paper, a new solid propellant microthruster design is proposed. The simulation to predict the microthruster performance is described. The fabrication and assembly of

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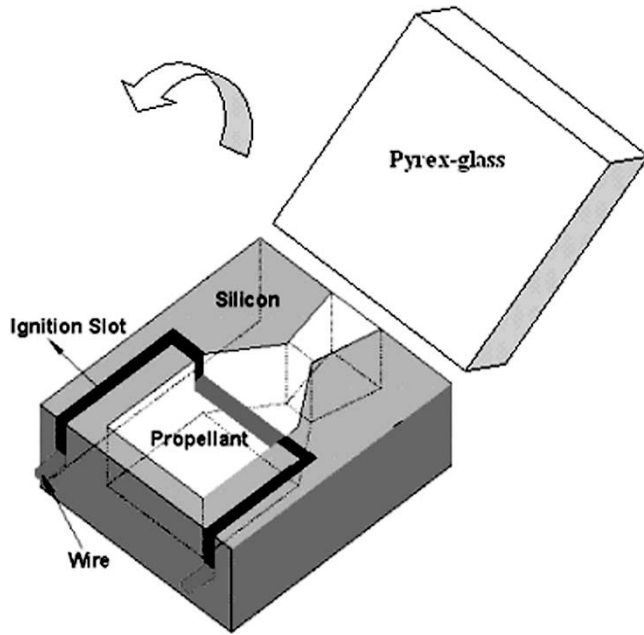


Fig. 1. Schematic of a single microthruster.

the microthruster are presented. Experiments to evaluate the feasibility of the new design are also discussed.

II. MICROTHRUSTER DESIGN

The proposed solid propellant microthruster configuration has no pumps, fuel lines and valves. Therefore there are no moving parts and the leakage possibility of the propellant is low. In the design, a silicon layer is fabricated to contain a combustion chamber, a convergent-divergent nozzle and an ignition slot. A specific glass layer is diced with the same dimensions as the silicon layer and is bonded together with the silicon layer to form a three-dimensional microthruster. The chamber is then loaded with the solid propellant. Once ignited, the resultant gas expands through the nozzle as its velocity increases drastically, thus producing the desired thrust and impulse bit. At the present study, a special ignition wire is installed in the ignition slot to ignite the propellant (with alternative ignition methods under consideration). The schematic view of a single microthruster is shown in Fig. 1.

In the design, the chamber, convergent-divergent nozzle and ignition slot are fabricated simultaneously. This makes the fabrication effective and efficient. Anodic bonding is employed here to bond the glass and silicon layers together, whose bonding quality is better than that of the bonding methods (cyanoacrylate adhesive, thermal epoxy, etc.) adopted by former researches. Normally, for the previous solid propellant microthruster designs, the combustion chambers and nozzles are etched vertically from the wafer surface. Consequently, longer chamber and nozzle lengths are difficult to fabricate due to the depth limitation of microfabrication technologies. Furthermore, the nozzle divergence angle and throat dimension have great impact on the performance of the microthruster. It is tough for the former designs to realize arbitrary nozzle divergence angle and throat dimension. Especially, KOH etching is sometimes employed to fabricate the nozzle. KOH etches silicon selectively

along the $\langle 100 \rangle$ planes, while almost stopping on the $\langle 111 \rangle$ planes, resulting in a fixed 35.3° angle between the nozzle edge and the centerline. However, for our design, the chamber and convergent-divergent nozzle can be etched horizontally along the wafer surface. Consequently, longer chambers and arbitrary nozzle dimensions can be fabricated according to the applications.

III. MODELING AND SIMULATION OF THE MICROTHRUSTER

A key point to the development of micropropulsion systems lies in the generation of extremely accurate thrust levels and impulse bits. Modeling and simulation can predict the processes inside the microthruster during combustion and the expansion of the gas through the nozzle and then derive the microthruster performance. Once the microthruster performance is predicted, the optimal design parameters can be determined. Some modeling ways to forecast the performance of microthruster have been presented [4], [9]–[11]. Navier–Stokes (N–S) equations-based modeling is undertaken here to establish a performance benchmark for the solid propellant microthruster. The combustion of the propellant is combined to the model as the inlet boundary condition. It is assumed that the pressure and temperature conditions in the combustion chamber are homogeneous. The species created from propellant combustion are all gaseous. The chamber temperature in the steady state is assumed to be 2000 K for the present propellant. Owing to the high chamber temperature, all the combustion gases are well above their saturation conditions and follow the ideal gas law.

A. Microthruster Geometry and Propellant Characteristics

The dimensions of the simulated single microthruster for the present study are shown in Fig. 2. The thickness of the wafer processed here is $650 \mu\text{m}$. The resulted trench depth is $350 \mu\text{m}$. The $100 \mu\text{m}$ plane in the nozzle throat is to avoid sharp edges and facilitate the fabrication. Moreover, the microthruster has different half divergence angle α and widths for its combustion chamber W_c , micronozzle throat W_t , and micronozzle exit W_e . We thus have different chamber-to-throat (A_c/A_t) and exit-to-throat (A_e/A_t) section ratios.

To achieve the smallest overall structure while preserving relatively high specific impulse, the composite solid propellant HTPB/AP/AL is chosen for the microthruster. The propellant normally contains 74% ammonium perchlorate (AP) oxidizer, 14% hydroxyl-terminated polybutadiene (HTPB) fuel binder, and other ingredients. Adding aluminum (Al) to the composite propellant is to increase the burn rate, flame temperature, and specific impulse. Table I lists the characteristics of the propellant [13].

B. Governing Equations

The mass conservation equation in Cartesian coordinates can be expressed as

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) = S_m \quad (1)$$

where the source item S_m denotes the mass added to the domain.

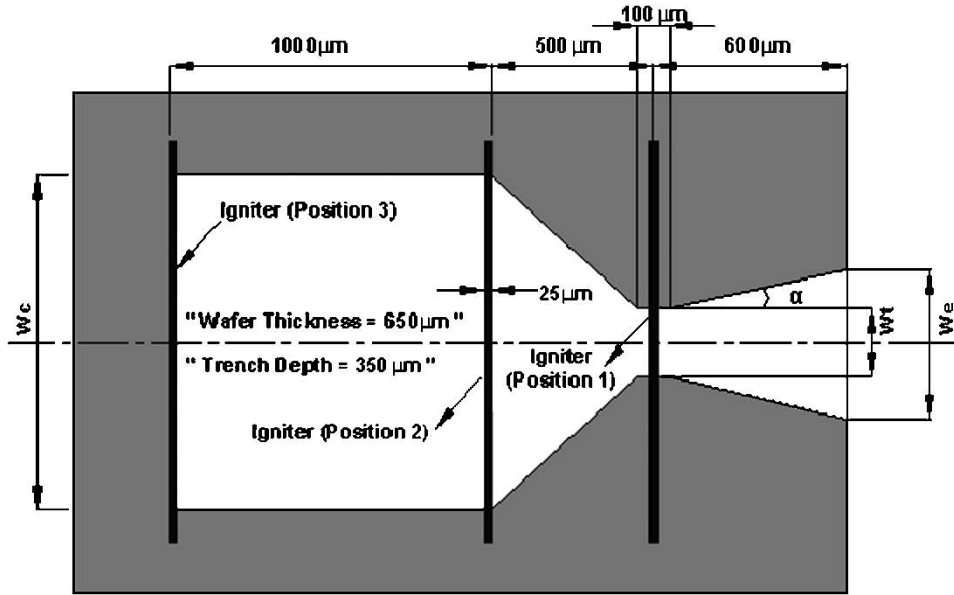


Fig. 2. Geometry of the solid propellant microthruster.

TABLE I
CHARACTERISTICS OF THE SOLID PROPELLANT

C_p (J·kg ⁻¹ ·K ⁻¹)	ρ_p (kg·m ⁻³)	a (kg ⁻ⁿ ·m ⁿ⁺¹ ·s ²ⁿ⁻¹)	n	γ (propellant gas)
1860	1854.6	6.35×10^{-5}	0.4	1.3

The momentum conservation equations can be written as

$$\begin{aligned} \frac{\partial}{\partial t}(\rho u) + \frac{\partial}{\partial x}(\rho u u) + \frac{\partial}{\partial y}(\rho u v) \\ = -\frac{\partial P}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + F_x \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho v) + \frac{\partial}{\partial x}(\rho v u) + \frac{\partial}{\partial y}(\rho v v) \\ = -\frac{\partial P}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + F_y \end{aligned} \quad (3)$$

where P is the pressure, τ_{ij} are components of the stress tensor and F_i are the gravity and body forces in the component directions.

Finally, the energy conservation equation is written as

$$\begin{aligned} \frac{\partial}{\partial t}(\rho E) + \frac{\partial}{\partial x_i}(u_i(\rho E + p)) \\ = \frac{\partial}{\partial x_i} \left(k \frac{\partial T}{\partial x_i} - \sum_j h_j J_j + u_j \tau_{ij} \right) + S_h \end{aligned} \quad (4)$$

where E is the total energy per unit mass, J_j is the diffusion flux of species j and x_i denotes summation over $i = 1, 2, 3, \dots$. The source item S_h denotes heat of chemical reaction and any other volumetric heat sources.

C. Boundary Conditions

The control volume should extend well beyond the microthruster itself, so that the domain boundary will not influence the problem unrealistically. Fig. 3 shows the control

volume and the boundary conditions for the microthruster model.

1) *Inlet Condition:* Disregarding the in-stationary effects of pressure on the propellant burning rate, the classical law (5) is used to determine the burning rate for the noncorrosive combustion of the propellant [13].

$$r = aP_c^n \quad (5)$$

where r is the burning rate that is defined as the spatial rate of change of the burning surface normal to the propellant surface. a (temperature coefficient) is an empirical constant influenced by the propellant grain temperature before burning. The rate exponent n , called the combustion index, is independent of the initial propellant temperature and describes the influence of chamber pressure on the burning rate.

The classical law (5) expresses the fact that the burning rate is dependent on the propellant composition and the chamber pressure of combustion. The chamber pressure, however, is determined by the equilibrium that should exist between the gas generation rate and the nozzle exhaust flow rate. Thus, the stable pressure level that is achieved is determined by the propellant composition and the ratio between the propellant burning area A_c and the nozzle throat area A_t . The micronozzle entrance (base) consists of the burning solid propellant with a surface burning rate per unit area given by

$$\frac{\dot{m}}{A_c} = \rho_p r = \rho_p a P_c^n. \quad (6)$$

2) *Wall Condition:* Heat flux through the wall at the inlet is assumed to be zero. All other walls are set to be two-sided walls. That means fluid regions exist on one side and solid regions exist on the other side of these walls. The two-sides walls have coupled thermal boundary conditions. The material of the wall is silicon. The silicon properties are: thermal conductivity = $141.2 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$; density = $2330 \text{ kg} \cdot \text{m}^{-3}$; heat capacity = $700 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$.

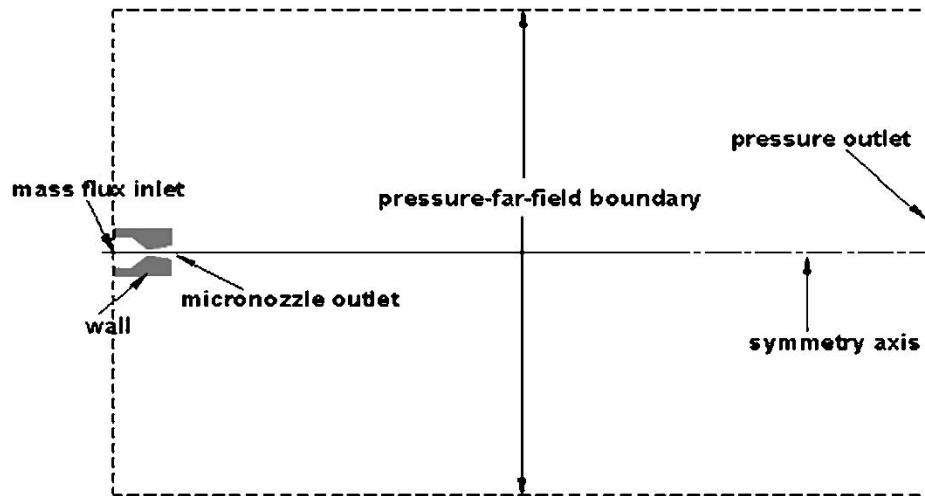


Fig. 3. Control volume and boundary conditions.

Knudsen number (Kn), the ratio of mean free path (λ) to a characteristic length scale of the system, is used to describe various flow regimes. By computing the values of Kn throughout the flow field at the normal wall boundary condition of no-slip, it is found that Kn is between 0.001 and 0.1. This suggests the need to consider velocity slip. One popular approach is to apply the N-S equations with a slip-flow boundary condition. In the present study, the wall is assumed to be stationary. The wall velocity, which is previously assumed to be zero in the state vector, can be replaced by the term $(2 - \sigma_V / \sigma_V) \lambda (du/dy)_w$ [14]. Here, σ_V is assumed to be 1 due to higher wall roughness.

3) *Outlet, Far-Field, and Initial Conditions:* When the altitude equals zero (sea level), the outlet, far-field and initial conditions are: static air; $T = 300$ K; $P = 1$ atm; $\rho = 1.225$ kg $\cdot$ m $^{-3}$; $\mu = 1.7894 \times 10^{-5}$ kg $\cdot$ m $^{-1} \cdot$ s $^{-1}$; $k = 0.0242$ W $\cdot$ m $^{-1} \cdot$ K $^{-1}$. When the altitude equals 100 000 m (space), the outlet and far-field conditions are: static air; $T = 195.08$ K; $P = 3.1593 \times 10^{-7}$ atm; $\rho = 5.604 \times 10^{-7}$ kg $\cdot$ m $^{-3}$; $\mu = 1.7894 \times 10^{-5}$ kg $\cdot$ m $^{-1} \cdot$ s $^{-1}$; $k = 0.0242$ W $\cdot$ m $^{-1} \cdot$ K $^{-1}$ [13].

D. Computation

Across the defined control volume (see Fig. 3), the N-S equations are solved using a finite-element software-FLUENT [15], which has been experimentally validated as a powerful CFD code for solving microcombustion and microflow problems [12]. FLUENT allows users to apply boundary conditions as user defined functions. Both the inlet boundary condition and the slip-flow wall boundary condition are programmed in C language and inserted into FLUENT as user defined functions. Fig. 4 illustrates a typical computational grid used for the numerical model. A high grid density is used near the nozzle walls, throat and exit to resolve the boundary layers and capture the possible shocks and expansion waves. The domain is meshed using 19 800 cells. The number of cells is determined by performing grid independent study to minimize the modeling error.

E. Computational Results

For the initial numerical simulations, both turbulent and laminar flow modeling are performed for the microthruster.

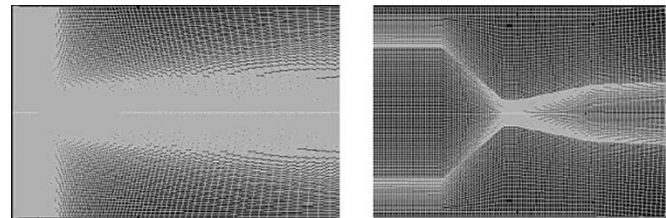


Fig. 4. Typical computational grid of numerical model with enlarged section of the nozzle.

The resulted flows experience Reynolds number below 2000. Therefore, the laminar flow modeling is safely employed for the following simulations. Given the nature of the gas expansion through the nozzle, the flow is also compressible. A three-dimensional simulation would provide a complete picture of the flow and an accurate performance prediction. However, this will come at a high computational price. A two-dimensional simulation will provide an accurate performance prediction for our cases in which the endwall boundary layers are thin with respect to the trench depth. The endwalls mean the upper and bottom walls of the microthruster. The computational results for steady-state are shown in following sections.

1) *Effect of Wall Condition on Performance:* In order to verify the proper formulation for wall conditions, the performances for zero thermal resistance wall, silicon wall, adiabatic wall, no-slip wall and slip wall are compared. Fig. 5 shows the effect of wall conditions on the axis pressure for the microthruster with $Wc = 1000$ μ m, $Wt = 300$ μ m, half divergence angle $\alpha = 16^\circ$. In the abscissa, 0 corresponds to the position inside the chamber 500 μ m away from the end of the chamber; 0.001 corresponds to the center of the nozzle throat plane; 0.0016 corresponds to the nozzle exit. Both at sea level and in space, the chamber pressure for adiabatic wall is higher than that of zero thermal resistance wall and silicon wall. The reason is that the thermal loss through the wall is neglected if the microthruster wall is assumed to be adiabatic. However, the chamber pressures for zero thermal resistance wall and silicon wall are almost the same both at sea level and in space. This is due to the high thermal conductivity of silicon comparing to the combustion gas and ambient air. It can be seen that

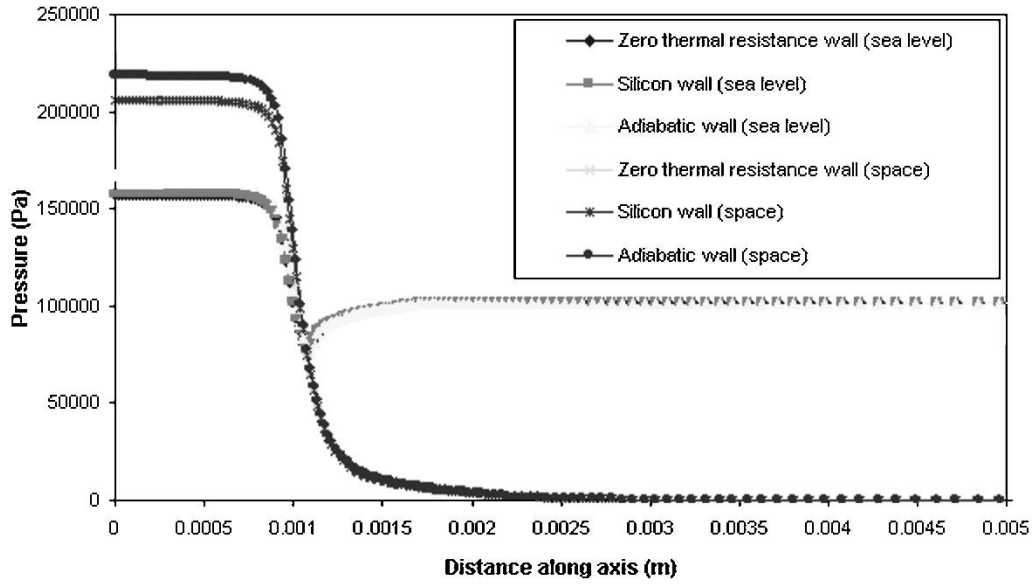


Fig. 5. Pressure profiles along the axis at different wall conditions.

TABLE II
THRUST VARIATIONS WITH WALL CONDITIONS

	Zero thermal resistance wall	Silicon wall	Adiabatic wall	Silicon wall with no-slip condition	Silicon wall with slip condition
Thrust (mN) Sea level	1.7699	1.7713	2.0928	1.6944	1.7713
Difference	0.079 %			4.341%	
	15.362%				
Thrust (mN) In space	13.1401	13.1461	15.2068	12.6436	13.1461
Difference	0.046 %			3.822 %	
	13.551 %				

the chamber pressure is different for sea level and in space. Depending on the nozzle throat geometry and the flow regime, whether subsonic or supersonic, the mass flow through the nozzle and the chamber pressure will be different. The ambient temperature difference is another cause although the difference (104.92 K) is small comparing to the combustion temperature (2000 K).

Table II records the performance variations for different wall conditions. At sea level, the thrust difference between zero thermal resistance wall and silicon wall is 0.079%, which is very small. However, the thrust differences of adiabatic wall and silicon wall, no-slip silicon wall and slip silicon wall are 15.362% and 4.341%, respectively. In space, the differences are 0.046%, 13.551%, and 3.822%, respectively. It is well known that when adiabatic wall and nonslip wall conditions are assumed, simulation results differ from actual situations. Therefore, the heat loss through the wall should not be ignored and the slip wall condition should be employed. The effect to the thrust of the silicon thermal resistance is very small. However, to improve the simulation precision, the silicon thermal resistance is also considered for the following simulations.

2) *Flow Field Analysis*: Fig. 6 shows the Mach number (the ratio of the gas velocity to the sound speed in the air) contours of a typical microthruster in space. The subsonic flow

accelerates out of the chamber through the convergent section, reaching sound speed at the throat (Mach number = 1). The flow through the nozzle is now choked. In the divergent part, the supersonic flow accelerates as the area gets bigger and exhausts as a supersonic jet.

Fig. 7 shows the Mach number profiles along the axis at different nozzle throat widths (Wt) for the microthruster with $Wc = 1000 \mu\text{m}$ and half divergence angle $\alpha = 12^\circ$. At sea level, when $Wt = 400 \mu\text{m}$ and $300 \mu\text{m}$, the flow through the nozzle is completely subsonic (i.e. the nozzle is not choked). The flow accelerates out of the chamber through the convergent section, reaching its maximum (subsonic) velocity at the throat. The flow then decelerates through the divergent section and exhausts into the ambient as a subsonic jet. When $Wt = 260 \mu\text{m}$, the flow pattern is exactly the same as in subsonic flow, except that the flow velocity at the throat has just reached Mach 1. Flow through the nozzle is just choked. When $Wt = 200 \mu\text{m}$, the Mach number at the throat is still 1. But a region of supersonic flow forms just downstream of the throat. Unlike a subsonic flow, the supersonic flow accelerates as the area gets bigger. This region of supersonic acceleration is terminated by a normal shock wave. The shock wave produces a near-instantaneous deceleration of the flow to subsonic velocity. This subsonic flow then decelerates through the remainder of the divergent section

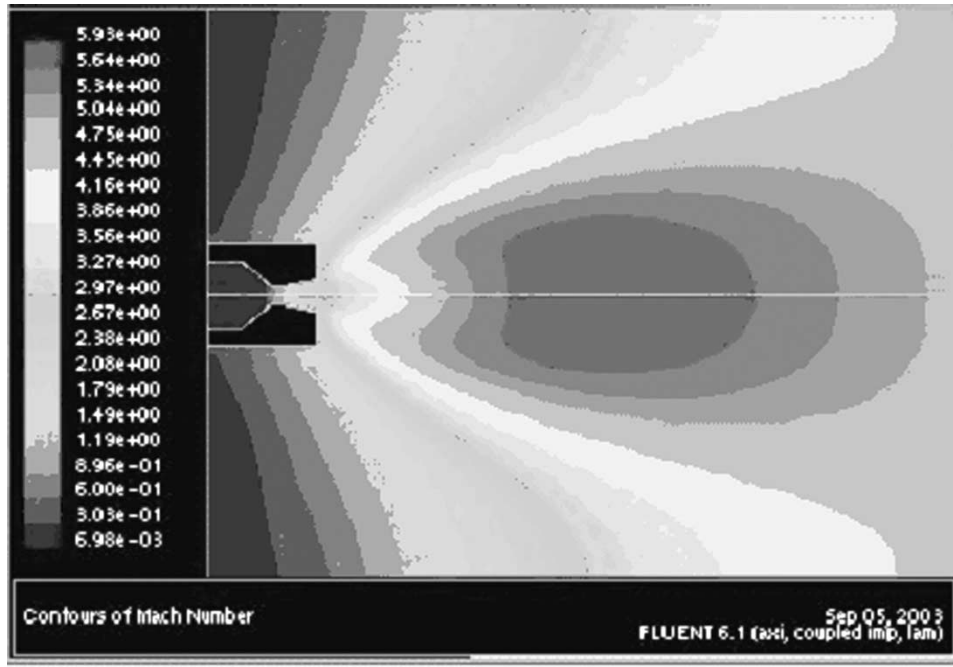


Fig. 6. Contours of Mach number.

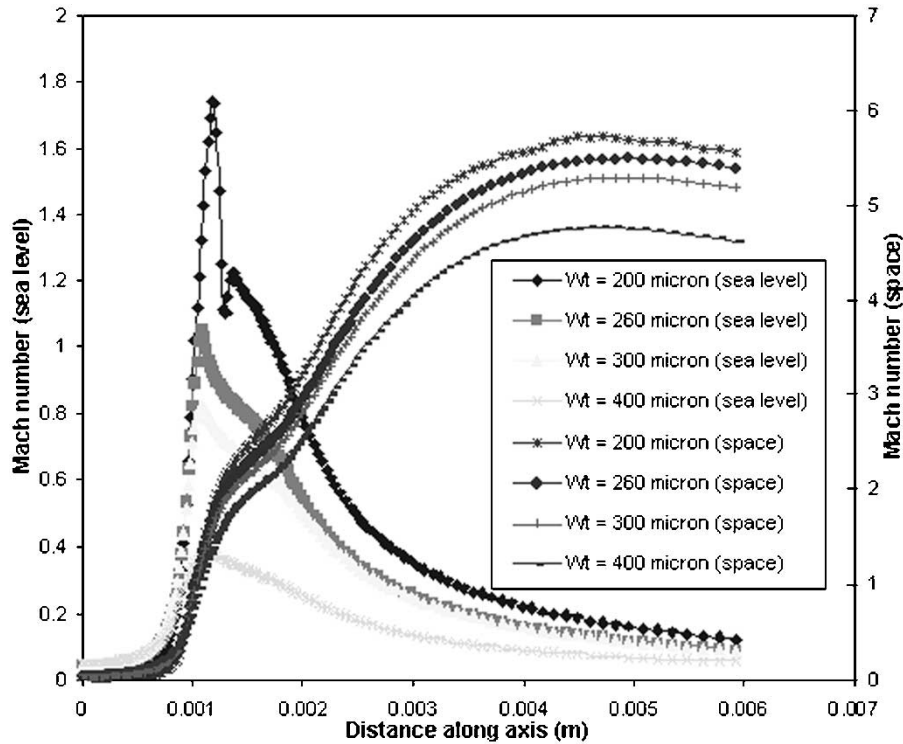


Fig. 7. Mach number profiles along the axis at different nozzle throat widths.

and exhausts as a subsonic jet. In space, the flow patterns for different nozzle throat widths are similar. The flow in the chamber and convergent section is subsonic. Then it reaches sonic flow at the throat (Mach number = 1). In the divergent part, the supersonic flow accelerates when the area gets bigger and exhausts as a supersonic jet.

3) *Thrust and Impulse of Microthruster*: Subsequent to combustion of the propellant, the expansion of the gas through

the nozzle produces a thrust T and total impulse I_t . The expression for thrust T is given as

$$T = \dot{m}V_e + (P_e - P_a)A_e. \quad (7)$$

The values of V_e and P_e are obtained by averaging the integration of velocity and pressure along the nozzle exit. Total impulse

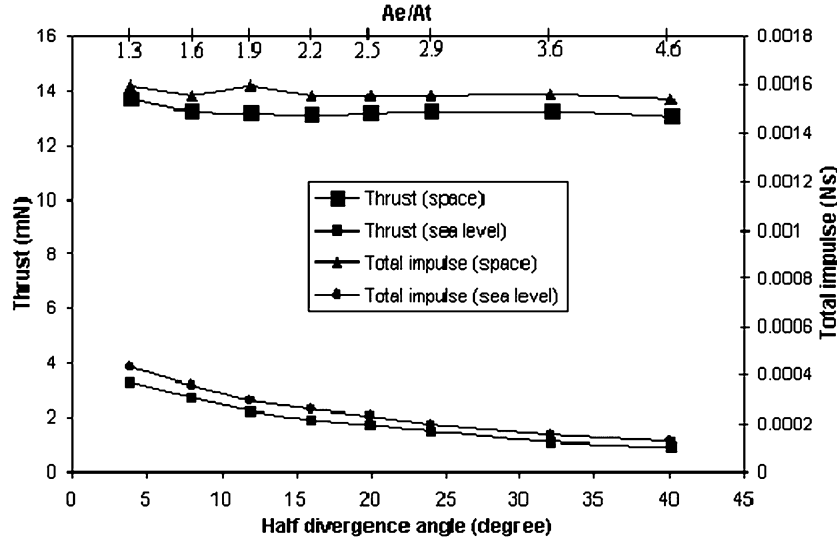


Fig. 8. Thrust and total impulse variations with the half divergence angle and A_e/A_t ratio.

I_t is another important parameter to describe the propulsion performance. I_t is the thrust T integrated over the burning time. For steady state, thrust T is constant. The burning time is the function of chamber length and burning rate. The chamber length is 1 mm for the designed microthrusters. The burning rate r is calculated from $r = aP_c^n$, where chamber pressure P_c is obtained from the simulation.

The half divergence angle (α) of the nozzle will impact the gas expansion process. As a result, it will also impact the thrust and impulse levels. To illustrate and qualify the impact of α on the thrust and impulse, the gas flow is computed for the microthrusters with different α and $W_c = 1000 \mu\text{m}$, $W_t = 260 \mu\text{m}$. Fig. 8 shows the thrust and total impulse variations with α . At sea level, when α increases, the thrust and total impulse decrease slightly and tend to become stable around 1 mN and $1.26 \times 10^{-4} \text{ N}\cdot\text{s}$, respectively for $\alpha > 35^\circ$. When α is too big (for $\alpha > 20^\circ$ in our design), nozzle exit pressure is lower than ambient pressure. The nozzle is overexpanding. However, when α is too small (for $\alpha < 8^\circ$ in our design), the fluid expansion is incomplete. The nozzle exit pressure is higher than ambient pressure. The nozzle is underexpanding. Both the overexpanding and underexpanding will affect the nozzle performance in terms of thrust and impulse. Optimum expansion is obtained when the fluid exits at ambient pressure. In our design, the optimum expansions are achieved at $10^\circ < \alpha < 20^\circ$ at sea level. The half divergence angle (α) is proportional to the A_e/A_t ratio due to the constant values of nozzle divergence length ($600 \mu\text{m}$) and throat width ($260 \mu\text{m}$) shown in Fig. 8. The optimum expansions are achieved at $1.75 < A_e/A_t < 2.54$ at sea level. In space, the impact of α and A_e/A_t ratio to the thrust and total impulse is not very remarkable according to the simulation, which is mainly due to the short nozzle divergence length.

As far as the nozzle design is concerned, the A_e/A_t ratio is another important parameter that impacts the thrust and impulse levels. In our present design, the chamber and nozzle are realized by deep reactive ion etching (DRIE). The chamber and nozzle have the same depth ($350 \mu\text{m}$). Consequently, A_e/A_t ratio equals W_c/W_t . By decreasing W_t of the nozzle throat,

the chamber pressure is greatly increased and the gas can be accelerated to a higher velocity, which will result in a greater thrust level. Fig. 9 shows the variations of the thrust and total impulse as a function of A_e/A_t ratio for the microthrusters with $W_c = 1000 \mu\text{m}$ and half divergence angle $\alpha = 12^\circ$. At sea level, the thrusts range from 0.76 mN to 4.38 mN and the total impulses range from $1.16 \times 10^{-4} \text{ N}\cdot\text{s}$ to $3.62 \times 10^{-4} \text{ N}\cdot\text{s}$. One key to designing an efficient subsonic nozzle lies in adjusting the nozzle throat area in order to have a fluid throat velocity close to the sound speed. For our microthrusters, this is achieved for $A_e/A_t = 3.85$ at sea level. In space, the thrust increases from 9.11 mN to 26.92 mN when the A_e/A_t ratio changes from 2.5 to 10. However, the total impulse decreases from $1.70 \times 10^{-3} \text{ N}\cdot\text{s}$ to $1.25 \times 10^{-3} \text{ N}\cdot\text{s}$ when A_e/A_t ratio increases from 2.5 to 10. The reason is that not only the thrust increases but also the chamber pressure increases when the A_e/A_t ratio increases. The chamber pressure increase causes burning rate to rise, thus resulting in the decrease of the burning time. Therefore, as the integration of thrust and burning time, the total impulse decreases. In space, the jet is uniformly supersonic for the A_e/A_t ratios in our design. This situation, since it is often desirable, is referred to as the ‘design condition’. According to the results shown in Fig. 8 and Fig. 9, the thrust in space is higher than that at sea level for the same microthruster geometry. This is due to the supersonic flow and increase of exit-ambient pressure difference in space.

IV. MICROTHRUSTER FABRICATION

A. Two-Dimensional (2-D) Microthruster Fabrication

A double-polished 6-inch (100) oriented silicon wafer is cleaned. A $10\text{-}\mu\text{m}$ -thick positive photoresist is deposited by spin coating at 1000 rpm. Then the photoresist is patterned by exposure to ultraviolet (UV) light through a mask. The exposed photoresist is developed. With the pattern transferred through lithography, the DRIE is performed on the wafer. The DRIE process is a succession of SF_6 etch cycles and C_4F_8 passivation cycles performed in a STS etcher. When

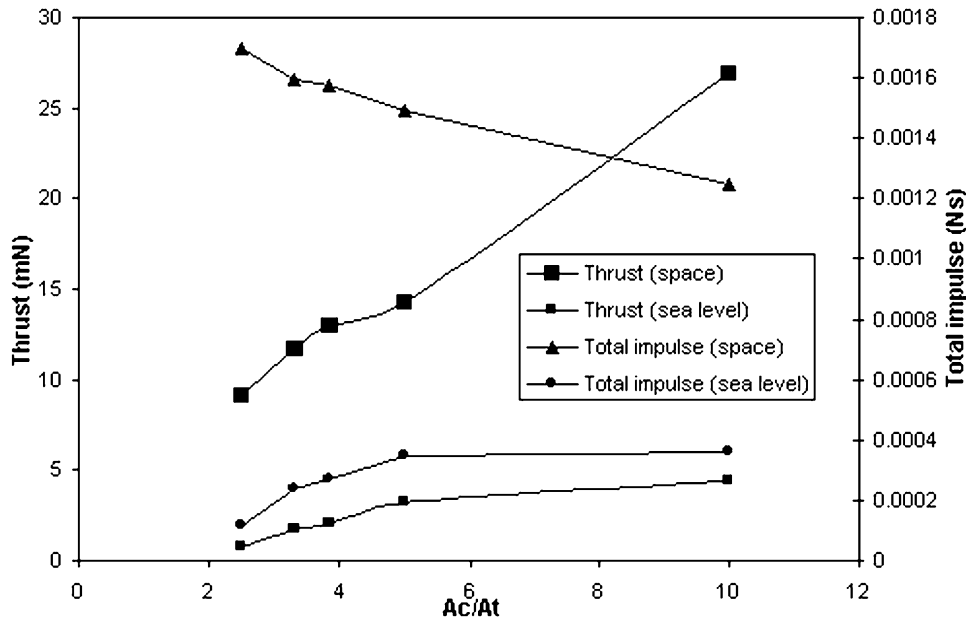


Fig. 9. Thrust and total impulse variations with the A_c/A_t ratio.

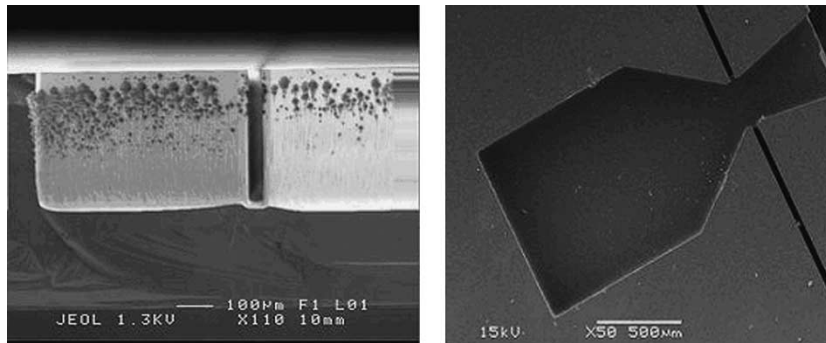


Fig. 10. SEMs of the cross-section and the front-side of the microthruster.

the photoresist is stripped and the wafer is cleaned, a thermal oxidation process is performed. The whole wafer is covered by the silicon dioxide with a thickness of 2180 Å after the thermal oxidation. The purpose of the thermal oxidation is to improve the insulation performance of the silicon wafer, which will improve the ignition efficiency by minimizing the current leakage. The resulted silicon trench depth is 350 μm as shown in Fig. 10. It can be seen that the sidewalls are vertical during and after the etching. Last, the individual microthrusters are separated using a dicing machine. Fig. 10 shows the scanned electron micrographs (SEMs) of a 2-D microthruster.

B. Ignitor Installation, Propellant Injection, and Three-Dimensional (3-D) Microthruster Formation

The wafer is diced into separate microthruster chips by a dicing machine. A Pyrex-7740 glass is diced with the same dimensions as the microthruster to form the seal. At the present study, a nickel chromium aluminum copper wire with a diameter of 25 μm is employed as the ignitor, which is installed in the 30-μm slot of the microthruster under a microscope. In our design, the ignitor can be placed in the throat, the middle or the back of the combustion chamber. The disadvantage of placing the ignitor on the back of the chamber is that portion

of the propellant will expel before combustion. When the ignitor is placed at the throat, the ignitor will affect the gas flow in the throat. Therefore, the ignitor should be placed at the middle of the chamber as a compromise. The optimal position can be evaluated by experiments. The microthruster with the ignitor is cleaned using ultrasonic agitation, rinsed in DI water and blown dry with a nitrogen gas. This is followed by an anodic bonding process at 400 °C and a voltage of 1100 V. A reliable bond between the silicon layer and glass layer is then formed although there is a 2180 Å silicon dioxide layer between the silicon and the glass. Finally the solid propellant powder is filled slowly into the microthruster in order to expel the trapped air inside the chamber. The fabricated 3-D microthruster with ignitor and solid propellant is shown in Fig. 11.

One disadvantage of the solid propellant microthruster is the lack of restart ability. However, the disadvantage can be partially redeemed by microthruster arrays. Even if some of the individual microthrusters fail to work, the array with some redundant microthrusters can still deliver the designed thrust and impulse bit. Furthermore the microthrusters in the array can be fired individually, several together or in controlled sequences, so as to produce controlled, vectored thrust according to the requirements. In our design, the microthruster array

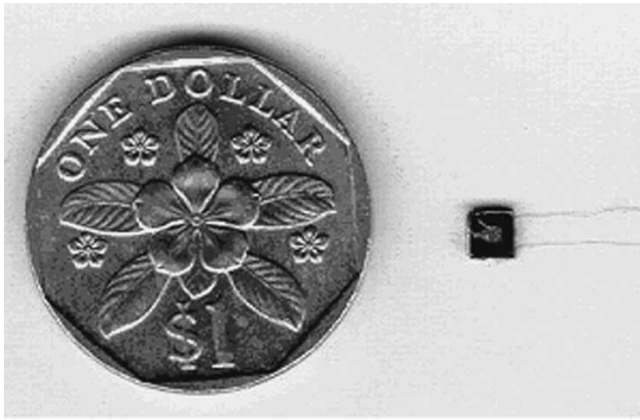


Fig. 11. Microthruster with propellant and ignitor.

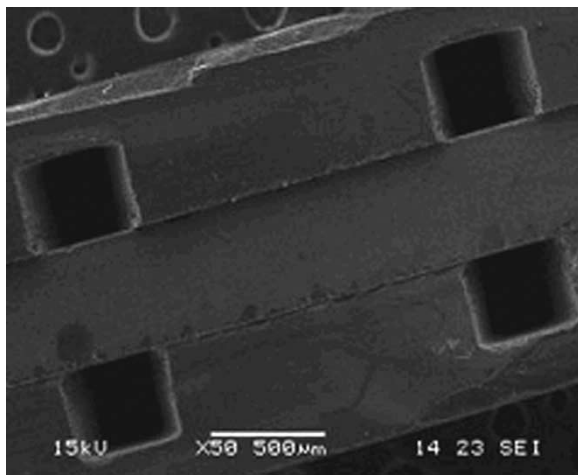


Fig. 12. SEM of microthruster array.

can be fabricated by bonding single microthruster and/or microthruster layer together. The bonding can be realized simply as building blocks for diverse ways to suit the application. One of the fabricated microthruster arrays is shown in Fig. 12. Adding addressing capability is one key part of developing the microthruster array. A schematic of one way to address and ignite single microthrusters individually in the microthruster array to produce the controlled, vectored thrust and impulse bit is shown in Fig. 13.

V. PRELIMINARY TESTING

Preliminary combustion testing has been performed to evaluate the feasibility of the new microthruster design. The increased surface area-to-volume ratio that comes with the small size of the microthruster can cause chemical and thermal quenching. The quenching can keep the microcombustion from occurring and propagating. Since the fabricated microthruster dimensions are small with a minimum nozzle feature of 100 μm , the quenching problem for the devices should be tested.

Continuous combustion is observed after igniting the solid propellant using the ignitor for the microthrusters with $W_c = 1000 \mu\text{m}$, $W_t = 260 \mu\text{m}$ and half divergence angle $\alpha = 12^\circ$. Successful production of thrust is verified by the microthruster displacement. A series of frames from a high-speed digital video

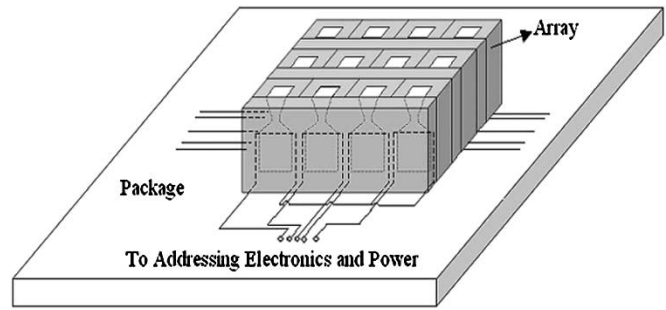


Fig. 13. Schematic of addressing and igniting single microthrusters in an array.

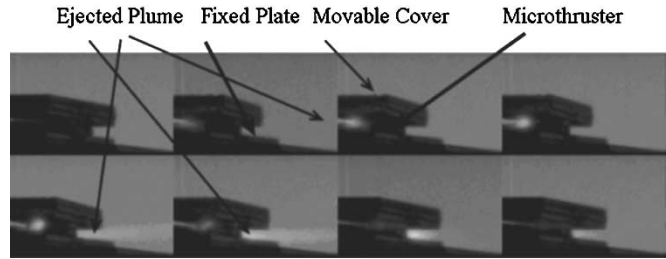


Fig. 14. Microthruster firing (images are acquired at 500 frames/s).

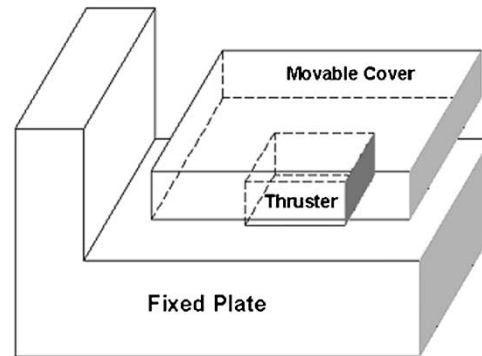


Fig. 15. Schematic of the geometry in Fig. 14.

camera of the microthruster experiment are shown in Fig. 14 and Fig. 16. The images shown in Fig. 14 are acquired at 500 frames/s. The ejected plume can be clearly seen from the series of images. The microthruster is placed between a fixed plate and a movable cover to stabilize the microthruster. The schematic diagram of the geometry in Fig. 14 is shown in Fig. 15. However, the cover is not heavy enough to prevent the microthruster from rotating owing to the produced thrust. The microthruster rotation is indicated by the variation of ejected plume direction. The thrust production can be proven by the microthruster rotation. Accurate measurement of the produced thrust and impulse bit will be performed in the future experiments. The images shown in Fig. 16 are acquired at 5000 frames/s, and they are taken in a different direction from those shown in Fig. 14. Two microthrusters are fired. The combustion occurs at different times after ignition.

Ignition occurs with input voltage of 14 V. Although current is not measured directly, the dc power supply current outputs of 68 mA for microthruster shown in Fig. 14 and 73 mA for microthrusters shown in Fig. 16 are observed at the time of ignition. The power supplied during ignition is then estimated to be 0.95 W and 1.02 W. According to the images, the duration

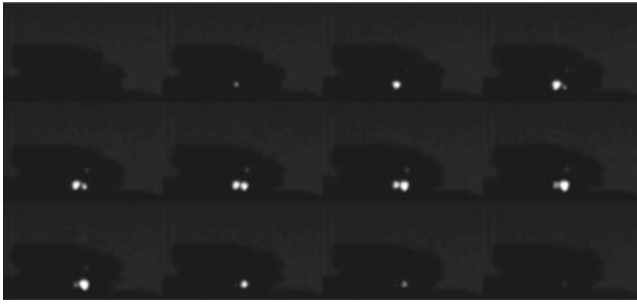


Fig. 16. Microthruster firing (images are acquired at 5000 frames/s).

of combustion is observed as 146 ms and 183 ms for the microthrusters shown in Fig. 14 and Fig. 16, respectively, which are longer than that of the estimated burning time (132.12 ms) through simulation. The duration difference is probably due to the combustion overlap (see Fig. 16), the neglect of the unsteady combustion after ignition, the nonuniformities of the propellant grain, or the heat loss.

VI. CONCLUSION AND FUTURE WORK

A MEMS-based solid propellant microthruster is considered as one of the ideal devices for meeting microspacecraft and micropropulsion requirements. Simulation has been performed to get the optimal microthruster configuration and to predict the performance. It is possible to vary the thrust and impulse depending on the application by adjusting geometrical parameters and propellant characteristics. The present prototype solid propellant microthruster is developed using microfabrication technologies. DRIE is employed to obtain the 2-D microthruster. The microthruster silicon wafer and Pyrex-7740 glass are diced into separate silicon and glass layers. Then anodic bonding is adopted to form 3-D microthruster. Preliminary combustion testing has been performed to evaluate the design feasibility. Continuous combustion has been achieved after igniting the propellant and successful production of thrust has been verified by the microthruster displacement.

Work is ongoing to measure the micro-scale thrust and impulse produced by the single microthrusters with diverse dimensions and microthruster arrays. The application of the classical burning rate law $r = aP_c^n$ in the micro-scale thruster has limitations. The experimental characteristic of this law customized to our microthruster is needed and will be explored next. The combustion gases are assumed to follow the ideal gas law, which should be studied further due to the aluminum contained in the propellant. Using nano aluminum instead of conventional aluminum will make the combustion gases much more closer to the idea gas. Nano aluminum burns rapidly and it is completely converted to oxide while still on the surface of the burning grain. The resulting oxides are aerosolized and reach thermal equilibrium with the combustion gases, and appear to follow the flow of the combustion gases in the slip stream. Various solid propellants, such as special gunpowder, glycidyle azide polymer (GAP) based propellant, and propellant of nano

size grain will be evaluated for their application in the solid propellant microthruster. The present wire ignitor is not optimum for the ignition. Other kinds of ignitors that are suitable for batch fabrication are being studied. Addressing circuit (diode, transistor, MEMS switch, etc.) for the microthruster arrays will be designed. Microelectronics packaging, microrelays and microsignal processing units will need to be investigated for integration into an intelligent microthruster system.

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